CAD detail level = the program build (RPS tier) (v0.89.0)

The auto-CAD engine renders the look each build can afford: richer on capable hardware, lightweight (and numpy-free) on legacy.

LEGACY build → v1

- flat diffuse shading
- head-down (original projection)
- no colour parse, NO TEXTURE
- no numpy / no GPU needed
- runs on Win 7 / Vista, low-RAM

cache: <nsn>_v1.png (per-tier, separate)

LITE build → v2

- right-side-up orientation
- Blinn-Phong specular + metallic
- no surface texture (lighter)
- good on mid machines
- numpy optional

cache: <nsn>_v2.png (per-tier, separate)

MODERN build → v3

- + FLIS colour (olive drab, CARC...)
- + per-material surface texture
- (brushed / grain / rubber / paint)
- the full 'product render'
- the default

cache: <nsn>_v3.png (per-tier, separate)

How /cadimg picks the level

Explicit ?style=v1|v2|v3 wins; else ?tier=modern|lite|legacy; else the server's own RPS_MODE (set from the machine profile at startup). cad_render.TIER_STYLE maps tier→style. Response carries X-CAD-Style. So a build serves — and only renders/ships — the detail it can afford; nothing wasted.

Populate a tier

MAKE-CAD.bat (or make_cad.py --style v1|v2|v3) pre-renders a chosen tier into the per-tier cache; the legacy build renders v1 on-demand with zero extra dependencies. Backwards-compatible: default stays v3 (R1).

Diagram. Dark (R3). v0.89.0 · 2026-06-04 · cad_render.TIER_STYLE + ensure(style=) · /cadimg ?style/?tier/RPS_MODE · make_cad --style. R1.